Nourhan Aly

Computer Science Student | Software Engineering Intern Candidate
Bachelor of Science in Computer Science — University of the People | Expected Graduation: 2027–2028 | GPA: 4.0
Honors: Dean's List, 2025–2026 — recognized by University of the People for outstanding academic performance
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PROFESSIONAL SUMMARY

Computer Science student graduating 2027–2028 with hands-on experience designing, developing, testing, and troubleshooting Java and Python software, from language fundamentals through object-oriented design, data structures, and small applications. I am at an intermediate level and almost done with Spring Boot, Postman, and Git, including REST APIs in Java, endpoint testing, and small commits from the terminal. Writes secure, well-documented code following established best practices, communicates clearly in writing and verbally, and collaborates effectively with others to solve problems and deliver reliable solutions. Comfortable using approved AI-assisted development tools responsibly. Seeking a Software Engineering internship on a fast-paced, collaborative engineering team.

TECHNICAL SKILLS

Languages: Java, Python, SQL
Core Concepts: Object-oriented programming, data structures, algorithms, interfaces and polymorphism, input validation, debugging
Libraries and Tools: Spring Boot, Postman, Git, Java Swing, Java Streams, Pandas, NumPy, Matplotlib, HTML, CSS, JavaScript

TECHNICAL PROJECTS

Library Database — SQL (SQLite)
- Designed a three-table schema (Books, Members, Loans) with primary and foreign keys, a unique email, and CHECK constraints (quantity can't go below zero, return date can't precede loan date)
- Wrote a join, an update, and a delete in a script that rebuilds the tables and reruns cleanly from scratch

Library Catalog — Java
- Built a menu-driven app to add, remove, search, and view books, DVDs, and magazines, using a generic Catalog<T>, inheritance, and checked exceptions for duplicate and missing items

Team Course Project — Inventory Management System *Java, OOP, team of 4*
- Collaborated with three classmates to design, develop, and test a Java inventory-tracking application, dividing work across data modeling, core logic, and testing, and merging everyone's contributions into a working solution

Student Management System *Java, Swing, ArrayList, HashMap*
- Designed and developed a desktop GUI application to add, update, and track students and course grades

To-Do List Web App *HTML, CSS, JavaScript*
- Built a task app where users add, edit in place, complete, and remove tasks, with blank-input handling and localStorage so tasks persist after a refresh

Employee Data Analysis *Python, Pandas, Matplotlib*
- Read employee records from Excel into Pandas DataFrames, troubleshot data inconsistencies, and built line, bar, scatter, and pie charts to summarize the data

Full project index: github.com/NouraAly1/java-learning-projects · github.com/NouraAly1/SQL-PROJECTS · github.com/NouraAly1/web_projects · github.com/NouraAly1/My_Journey_Learning_Python

EXPERIENCE

STEM Programming Instructor — UNESCO Center for Peace Summer Camp *June 2026, Team Lead*
- Selected and prepared introductory programming and technology topics for students interested in STEM during a 3-week program
- Taught programming fundamentals to camp attendees, explaining technical concepts clearly for a range of skill levels

Teller (Part-time) — Fifth Third Bank *2023 – Present*
- Process customer transactions accurately while following bank compliance and security procedures
- Explain account details and procedures clearly to customers and coworkers to resolve requests

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networking, Introduction to Computer Science
Anthropic Academy (in progress): AI Fluency: Framework & Foundations, Claude Code 101, Building with the Claude API

CERTIFICATIONS

Python for Everybody — Coursera · View credential

LANGUAGES

English, Arabic, Japanese